

April – Four-Week Internship Project Plan

Four-Week Internship Project Plan

Intern: Yiyou “April” Qian

Position: AI-Consulting Analyst Intern

Company: Gateway Solutions

Project Duration: Four Weeks

Project Title: NYC Restaurant Intelligence Platform

1. Project Overview

The NYC Restaurant Intelligence Platform is a data analytics and decision-support product designed for Gateway Solutions and its restaurant-industry clients.

The platform will consolidate publicly available data into a structured and maintainable restaurant database. It will provide analytical tools for evaluating market conditions, identifying potential locations, measuring competitive intensity, assessing operational risks, and monitoring industry trends.

The intern will first establish the core database and application architecture. Following completion of the minimum viable product, additional analytical modules will be developed according to priority and available time.

2. Project Objectives

The primary objectives of the project are to:

- Develop a structured and cumulative database of New York City restaurants.
- Establish a repeatable process for collecting, cleaning, validating, and updating public data.
- Create transparent analytical frameworks for restaurant market evaluation.
- Develop a customer-facing web application for Gateway Solutions.
- Translate analytical findings into practical business recommendations.
- Produce technical and operational documentation to support future development.

3. Intended Users

The platform is intended to support:

- Restaurant owners evaluating new locations
- Restaurant groups considering expansion opportunities
- Entrepreneurs researching restaurant concepts

- Existing operators assessing their competitive position
- Gateway Solutions consultants preparing client analyses and proposals

4. Proposed Data Sources

The project will primarily use publicly available information from:

- NYC Restaurant Inspection Data
- NYC Open Data
- U.S. Census American Community Survey
- MTA Open Data
- NYC geographic boundary datasets
- NYC sidewalk café datasets
- NOAA weather and climate data

Each source must be documented with its URL, relevant fields, update frequency, retrieval date, and usage limitations.

5. Proposed Technology

The anticipated technical stack includes:

- Python
- Pandas and NumPy
- SQL
- SQLite or Supabase/PostgreSQL
- GeoPandas
- Plotly
- Streamlit
- scikit-learn, where appropriate
- GitHub for version control and documentation

Week 1: Data Infrastructure and Market Overview

Weekly Objective

Establish the project's analytical foundation by defining the business requirements, designing the central database, and developing an initial restaurant market overview.

Assigned Responsibilities

- Define the platform's intended users and principal business questions.

- Identify and evaluate the required public datasets.
- Prepare a formal inventory of all proposed data sources.
- Define the key performance indicators and analytical variables.
- Clean and standardize restaurant names, addresses, cuisine classifications, geographic fields, and dates.
- Establish a methodology for assigning unique restaurant identifiers.
- Design the database schema and table relationships.
- Load the initial cleaned data into SQLite or Supabase.
- Calculate restaurant counts by borough, neighborhood, ZIP code, and cuisine.
- Develop preliminary charts and geographic visualizations.
- Create the first version of the Streamlit Market Overview module.
- Document data-quality issues, assumptions, and limitations.

Required Deliverables

- Project requirements document
- Data-source inventory
- Data dictionary
- Database schema
- Initial cleaned restaurant dataset
- Data-cleaning and validation notebook
- Initial central database
- Market Overview minimum viable product
- Week 1 progress report

Expected Weekly Outcome

By the conclusion of Week 1, users should be able to explore the distribution of New York City restaurants by location and cuisine through an initial functional dashboard.

Week 2: Location and Competitive Market Analysis

Weekly Objective

Develop analytical tools that support restaurant location assessment and competitive-market evaluation.

Assigned Responsibilities

- Connect restaurant records with neighborhood, ZIP code, and Census information.
- Incorporate relevant population, household, income, and demographic variables.

- Calculate restaurant density by geographic area.
- Measure same-cuisine and overall restaurant competition.
- Develop an explainable Competitive Saturation Score.
- Identify potentially underserved cuisine and neighborhood combinations.
- Develop an initial Location Opportunity Score.
- Define and document the methodology and weights used in each score.
- Allow users to adjust selected assumptions or scoring weights where practical.
- Add filters for cuisine, borough, neighborhood, and relevant customer characteristics.
- Develop an interactive location opportunity map.
- Test the model using a selection of sample restaurant concepts.
- Review whether the resulting recommendations are logically consistent.

Required Deliverables

- Neighborhood-level analytical dataset
- Restaurant-density metrics
- Competitive Saturation Score
- Location Opportunity Score
- Location Opportunity Analyzer
- Competition and Market Gap Analyzer
- Scoring methodology documentation
- Week 2 progress report

Expected Weekly Outcome

By the conclusion of Week 2, a user should be able to select a cuisine and target market and receive a ranked list of potential neighborhoods with a clear explanation of the underlying results.

Week 3: Operational Risk and Additional Analytical Modules

Weekly Objective

Expand the platform to include restaurant operational-risk analysis and additional market intelligence capabilities.

Primary Assignment: Inspection Risk Analyzer

The intern will:

- Analyze restaurant inspection scores, grades, and violation histories.

- Develop categories for common operational violations.
- Identify repeated and potentially critical violations.
- Compare inspection patterns across cuisines, boroughs, and neighborhoods.
- Evaluate factors associated with unfavorable inspection outcomes.
- Develop an explainable Restaurant Inspection Risk Score.
- Create a general inspection-readiness checklist.
- Document the limitations of using historical inspection data to assess future risk.

Additional Modules

Following completion of the Inspection Risk Analyzer, the intern should develop additional modules in the following order of priority.

Restaurant Comparison Tool

The tool should allow users to compare restaurants or neighborhoods across:

- Cuisine and market characteristics
- Competitive intensity
- Demographic indicators
- Inspection performance
- Location opportunity
- Operational-risk indicators

Restaurant Trend Tracker

The tracker should analyze:

- Restaurant activity over time
- Cuisine growth and decline
- Neighborhood-level changes
- Inspection-grade trends
- Apparent restaurant market entries and exits

Outdoor Dining Opportunity Analyzer

If time permits, this module should evaluate:

- Sidewalk café information
- Restaurant locations
- Neighborhood characteristics
- Seasonal weather patterns
- Outdoor dining activity and potential

Required Deliverables

- Inspection Risk Analyzer
- Restaurant Inspection Risk Score

- Inspection-readiness checklist
- Restaurant Comparison Tool
- Restaurant Trend Tracker, if time permits
- Outdoor Dining Opportunity Analyzer prototype, if time permits
- Updated central database
- Model validation and limitation notes
- Week 3 progress report

Expected Weekly Outcome

By the conclusion of Week 3, the platform should contain at least three functional analytical modules supported by the same centralized database.

Week 4: Platform Integration and Final Handoff

Weekly Objective

Integrate the completed analytical modules into a coherent customer-facing product and prepare all materials required for continued use and development.

Assigned Responsibilities

- Consolidate the completed modules into a single Streamlit application.
- Establish clear and consistent platform navigation.
- Improve the presentation of filters, charts, maps, and scoring explanations.
- Add options for downloading relevant tables and analytical results.
- Develop a standardized customer summary report.
- Test the platform using multiple restaurant scenarios.
- Identify and correct data, calculation, and usability issues.
- Document the process for retrieving and updating each dataset.
- Prepare a formal data refresh standard operating procedure.
- Prepare a user guide for Gateway Solutions consultants.
- Organize the GitHub repository and ensure that the code is reproducible.
- Document incomplete features and recommended future enhancements.
- Deploy the application, if technically feasible.
- Prepare and deliver a formal final presentation and product demonstration.

Required Deliverables

- Final NYC Restaurant Intelligence Platform
- Cumulative and documented restaurant database

- At least three completed analytical modules
- Standardized customer summary report
- Organized GitHub repository
- README and installation instructions
- Database documentation
- Data refresh standard operating procedure
- Gateway Solutions consultant user guide
- Future product-development roadmap
- Final presentation and product demonstration
- Week 4 and final internship report

Expected Weekly Outcome

By the conclusion of Week 4, Gateway Solutions should receive a functional and reusable restaurant intelligence product that can be demonstrated to clients and continued by future interns or developers.

Project Priorities

Minimum Required Scope

The following components must be completed:

1. Central restaurant database
2. Restaurant Market Overview
3. Location Opportunity Analyzer
4. Competition and Market Gap Analyzer
5. Customer-facing Streamlit application
6. Technical and operational documentation

Secondary Priorities

The following components should be completed after the minimum required scope:

1. Inspection Risk Analyzer
2. Restaurant Comparison Tool
3. Restaurant Trend Tracker

Extended Scope

The following components should be treated as stretch objectives:

1. Outdoor Dining Opportunity Analyzer
2. Automated customer report generator

3. Scheduled database refresh
4. Production deployment
5. User account and saved-report functionality

Communication and Reporting Requirements

Tuesday Written Update

The intern will submit a concise written update addressing:

- Work completed since the previous meeting
- Preliminary findings
- Current deliverables
- Technical or analytical challenges
- Assistance required
- Planned work before the Thursday meeting

Thursday Progress Meeting

The weekly progress meeting will include:

- Demonstration of completed work
- Review of data quality and methodology
- Discussion of analytical findings
- Feedback on product functionality
- Resolution of outstanding issues
- Confirmation of priorities for the following week

Weekly Report

At the end of each week, the intern will submit a written report containing:

- Weekly objectives
- Work completed
- Deliverables produced
- Principal findings
- Challenges encountered
- Solutions implemented or proposed
- Skills and knowledge developed
- Priorities for the following week

Final Evaluation Criteria

The project will be evaluated according to:

- Accuracy and completeness of the collected data
- Quality of database design and documentation
- Effectiveness of data cleaning and validation
- Soundness and transparency of analytical methodologies
- Reproducibility and organization of the code
- Usability of the customer-facing application
- Relevance of the findings to restaurant clients
- Ability to convert analysis into practical recommendations
- Quality of technical and operational documentation
- Completeness of the final handoff
- Quality of the final presentation and product demonstration

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GATEWAY SOLUTIONS

FOUR-WEEK INTERNSHIP PROJECT PLAN

Project Information

Item	Details
Intern	Yiyou “April” Qian
Position	AI-Consulting Analyst Intern
Department	AI and Data Consulting
Project Title	NYC Restaurant Intelligence Platform
Project Duration	Four Weeks
Work Arrangement	Remote
Project Supervisor	Scott
Organization	Gateway Solutions

1. Project Background

Gateway Solutions provides consulting and product-development services to clients in the restaurant industry. Restaurant owners frequently require reliable market information when evaluating new locations, assessing competition, improving operational performance, and planning future expansion.

The purpose of this project is to develop a centralized restaurant intelligence platform using publicly available data. The platform will consolidate multiple datasets into a structured database and provide practical analytical tools for Gateway Solutions and its restaurant clients.

The project will emphasize data engineering, statistical analysis, operations research, business intelligence, and product development. Artificial intelligence may be used selectively to support specific functions, but it will not constitute the primary focus of the project.

2. Project Purpose

The NYC Restaurant Intelligence Platform will provide Gateway Solutions with a reusable decision-support product capable of addressing the following business questions:

1. Which New York City neighborhoods may present favorable opportunities for a particular restaurant concept?
2. How competitive is a selected neighborhood or cuisine category?
3. What operational and inspection risks are commonly associated with specific restaurant segments?
4. How are restaurant markets and cuisine categories changing over time?
5. How does a selected restaurant or neighborhood compare with relevant benchmarks?
6. What data-supported recommendations can Gateway Solutions provide to prospective restaurant clients?

3. Project Objectives

The intern will be responsible for pursuing the following objectives:

1. Develop a centralized and maintainable database of New York City restaurants.
2. Establish a repeatable process for collecting, cleaning, validating, and updating public data.
3. Develop transparent analytical methodologies for market, location, competition, and operational-risk evaluation.
4. Build a customer-facing web application through which users can explore the data and analytical results.
5. Translate quantitative findings into practical business recommendations.
6. Produce sufficient technical and operational documentation to support future use and development.
7. Complete additional analytical modules according to the established order of priority and available project time.

4. Project Scope

4.1 Required Components

The minimum required scope consists of the following components:

1. Central restaurant database
2. Restaurant Market Overview
3. Location Opportunity Analyzer
4. Competition and Market Gap Analyzer
5. Customer-facing Streamlit application
6. Data and methodology documentation
7. Data refresh standard operating procedure

4.2 Secondary Components

The following components should be developed following completion of the minimum required scope:

1. Inspection Risk Analyzer
2. Restaurant Comparison Tool
3. Restaurant Trend Tracker

4.3 Extended Components

The following components may be developed if project progress and available time permit:

1. Outdoor Dining Opportunity Analyzer
2. Automated customer report generator
3. Scheduled database updates
4. Public application deployment
5. User accounts and saved-report functionality

5. Proposed Data Sources

The project will primarily use publicly accessible information from the following sources:

Data Source	Proposed Use
NYC Restaurant Inspection Data	Restaurant records, grades, inspection scores, violations, and inspection history
NYC Open Data	Business, geographic, sidewalk café, and neighborhood
U.S. Census American Community Survey	Population, income, household, age, and demographic characteristics

MTA Open Data	Transportation access and relevant ridership indicators
NYC Geographic Boundary	Borough, ZIP code, Census tract, and neighborhood
NYC Sidewalk Café Data	Outdoor dining applications, status, location, tables, and
NOAA Weather Data	Temperature, precipitation, snowfall, and seasonal

For each data source, the intern must document:

- Source name and URL
- Data owner
- Relevant variables
- Retrieval date
- Update frequency
- Data-quality concerns
- Usage restrictions
- Refresh procedure

6. Proposed Technical Environment

Area	Proposed Technology
Programming Language	Python
Data Processing	Pandas and NumPy
Database	SQLite or Supabase/PostgreSQL
Database Connection	SQLAlchemy
Geospatial Analysis	GeoPandas
Visualization	Plotly
Web Application	Streamlit
Statistical Modeling	scikit-learn, where appropriate
Version Control	Git and GitHub
Documentation	README, data dictionary, methodology documentation, and user guide

7. Project Schedule

Week 1 — Data Infrastructure and Market Overview

Weekly Objective

Establish the analytical and technical foundation of the project by defining the business requirements, designing the central database, and developing an initial market overview.

Assigned Activities

1. Define the intended users of the platform.
2. Identify the principal business questions the platform must address.
3. Prepare an inventory of relevant public datasets.
4. Define the principal analytical variables and performance indicators.
5. Retrieve and inspect the initial restaurant datasets.
6. Clean and standardize restaurant names, addresses, cuisines, geographic fields, and dates.
7. Establish a methodology for assigning unique restaurant identifiers.
8. Design the database schema and table relationships.
9. Load the initial cleaned data into the selected database.
10. Calculate restaurant counts by borough, neighborhood, ZIP code, and cuisine.
11. Develop preliminary charts and geographic visualizations.
12. Create the initial Streamlit Market Overview page.
13. Document data assumptions, quality issues, and limitations.

Required Deliverables

Deliverable	Description
Project Requirements Document	Intended users, business questions, scope, and proposed functions
Data-Source Inventory	List and assessment of all proposed public datasets
Data Dictionary	Definition, type, source, and purpose of each principal variable
Database Schema	Tables, fields, primary keys, and relationships
Initial Restaurant Dataset	Cleaned and standardized restaurant-level data
Data-Cleaning Notebook	Reproducible data-processing and validation code
Central Database	Initial operational version of the cumulative database
Market Overview MVP	Functional first version of the dashboard
Week 1 Report	Summary of progress, findings, issues, and next steps

Expected Outcome

By the conclusion of Week 1, users should be able to explore the general distribution of New York City restaurants by location and cuisine through an initial working application.

Week 2 — Location and Competitive Market Analysis

Weekly Objective

Develop customer-facing analytical tools that support restaurant location selection and competitive-market evaluation.

Assigned Activities

1. Connect restaurant records with neighborhood, ZIP code, and Census information.
2. Incorporate relevant population, household, income, and demographic variables.
3. Calculate restaurant density by geographic area.
4. Measure overall and same-cuisine competitive concentration.
5. Develop an explainable Competitive Saturation Score.
6. Identify potentially underserved cuisine and neighborhood combinations.
7. Develop an initial Location Opportunity Score.
8. Define and document the variables and weights used in each score.
9. Provide users with adjustable scoring assumptions where practical.
10. Add filters for cuisine, borough, neighborhood, and target-market characteristics.
11. Develop an interactive location opportunity map.
12. Test the model using representative restaurant concepts.
13. Review whether the results are reasonable, stable, and explainable.

Required Deliverables

Deliverable	Description
Neighborhood Analytical Dataset	Consolidated restaurant, demographic, and geographic data
Restaurant Density Metrics	Restaurant concentration by location and cuisine
Competitive Saturation Score	Explainable measure of local competitive intensity
Location Opportunity Score	Multi-factor neighborhood opportunity assessment
Location Opportunity Analyzer	Customer-facing location evaluation module
Competition Analyzer	Market saturation and potential-gap analysis
Methodology Document	Variables, formulas, weights, assumptions, and limitations
Week 2 Report	Summary of progress, findings, issues, and next steps

Expected Outcome

By the conclusion of Week 2, a user should be able to define a restaurant concept and receive a ranked comparison of potential neighborhoods, together with an explanation of the results.

Week 3 — Operational Risk and Additional Analytics

Weekly Objective

Expand the platform to include operational-risk analysis and additional restaurant market-intelligence functions.

Primary Assignment: Inspection Risk Analyzer

The intern will:

- 1. Analyze restaurant inspection scores, grades, and violation histories.
- 2. Classify common operational violations.
- 3. Identify repeated and critical violations.
- 4. Compare inspection patterns across cuisines, boroughs, and neighborhoods.
- 5. Evaluate factors associated with unfavorable inspection outcomes.
- 6. Develop an explainable Restaurant Inspection Risk Score.
- 7. Create a general inspection-readiness checklist.
- 8. Document the analytical limitations of using historical inspection data to assess future risk.

Additional Assignments

Upon completion of the Inspection Risk Analyzer, the intern should develop the following modules in order of priority:

Prio rity	Module	Principal Function
1	Restaurant Comparison Tool	Compare restaurants or neighborhoods across market, demographic, competition, and inspection indicators
2	Restaurant Trend	Analyze changes in restaurant activity, cuisine categories,
3	Outdoor Dining Opportunity Analyzer	Evaluate outdoor dining potential using public location, café, geographic, and weather data

Required Deliverables

Deliverable	Description
Inspection Risk Analyzer	Operational-risk analysis module
Inspection Risk Score	Explainable restaurant risk assessment
Inspection-Readiness Checklist	General operational preparation framework
Restaurant Comparison Tool	Side-by-side restaurant or neighborhood analysis
Restaurant Trend Tracker	To be completed if time permits
Outdoor Dining Prototype	To be completed if time permits

Updated Central Database	Inclusion of all newly processed data
Validation and Limitation Notes	Evaluation of model performance and appropriate use
Week 3 Report	Summary of progress, findings, issues, and next steps

Expected Outcome

By the conclusion of Week 3, the platform should contain at least three functional analytical modules supported by a common database and consistent methodology.

Week 4 — Platform Integration and Final Handoff

Weekly Objective

Integrate all completed work into a coherent customer-facing product and prepare the materials necessary for future operation and development.

Assigned Activities

1. Integrate all completed modules into a single Streamlit application.
2. Establish a clear and consistent navigation structure.
3. Standardize filters, charts, maps, terminology, and scoring explanations.
4. Add downloadable analytical tables and results.
5. Develop a standardized customer summary report.
6. Test the platform using multiple restaurant scenarios.
7. Identify and correct data, calculation, and usability issues.
8. Document the retrieval and update process for each dataset.
9. Prepare a formal data refresh standard operating procedure.
10. Prepare a user guide for Gateway Solutions consultants.
11. Organize the GitHub repository and confirm code reproducibility.
12. Document incomplete functions and recommended future enhancements.
13. Deploy the application if technically feasible.
14. Prepare and deliver a formal final presentation and product demonstration.

Required Deliverables

Deliverable	Description
Final Platform	Integrated NYC Restaurant Intelligence Platform
Central Database	Cumulative, structured, and documented restaurant database
Analytical Modules	Minimum of three functional analytical modules

Customer Summary Report	Standardized client-facing analytical output
GitHub Repository	Organized code, data instructions, and version history
README	Installation, configuration, and usage instructions
Database Documentation	Schema, field definitions, and relationships
Data Refresh SOP	Repeatable procedure for updating all public data
Consultant User Guide	Instructions for Gateway Solutions personnel
Development Roadmap	Prioritized list of future product enhancements
Final Presentation	Formal project summary and product demonstration
Final Internship Report	Consolidated summary of work, findings, and recommendations

Expected Outcome

By the conclusion of Week 4, Gateway Solutions should receive a functional and reusable restaurant intelligence product that can be demonstrated to prospective clients and continued by future interns or developers.

8. Communication and Reporting Requirements

8.1 Tuesday Written Update

The intern will submit a written update each Tuesday containing:

1. Work completed since the previous update
2. Preliminary findings
3. Deliverables currently available for review
4. Technical or analytical challenges
5. Assistance or decisions required
6. Planned work before the Thursday meeting

8.2 Thursday Progress Meeting

The weekly meeting will include:

1. Demonstration of completed work
2. Review of data quality and analytical methodology
3. Discussion of findings and business implications

4. Feedback on product functionality and presentation
5. Resolution of outstanding questions or blockers
6. Confirmation of priorities for the following week

8.3 Weekly Report

A formal weekly report will be submitted at the conclusion of each week and will include:

1. Weekly objectives
2. Activities completed
3. Deliverables produced
4. Principal findings
5. Challenges encountered
6. Solutions implemented or proposed
7. Skills and knowledge developed
8. Priorities for the following week

9. Project Governance

The following principles will apply throughout the project:

- The central database and minimum viable product must be completed before extended modules are pursued.
- All calculations and scoring methodologies must be explainable and documented.
- Public data must be used in accordance with the applicable terms and limitations.
- Analytical results must be presented as decision-support indicators rather than guaranteed business outcomes.
- Code and documentation must be sufficiently organized for another team member to reproduce and continue the work.
- Project scope may be adjusted by the supervisor based on progress, data availability, and business priorities.

10. Final Evaluation Criteria

The intern's performance and final project will be evaluated according to the following criteria:

Evaluation Area	Standard
Data Quality	Accuracy, completeness, consistency, and appropriate validation
Database Design	Logical structure, maintainability, and documentation
Analytical Methodology	Sound reasoning, transparency, and appropriate limitations

Technical Quality	Reproducible, organized, and adequately documented code
Product Usability	Clear navigation, meaningful outputs, and customer relevance
Business Application	Ability to translate analysis into practical recommendations
Project Management	Timely delivery, prioritization, and responsiveness to feedback
Documentation	Completeness of technical, operational, and user materials
Communication	Clarity of written updates, meetings, and presentations
Final Handoff	Readiness for continued use and future development